Cohort study

Cohort study : Using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for analytical cohort study (last amended in 2023)

Study title (authors)	Major Components	Response options			
38 The relationship between myopia and near work, time outdoors and socioeconomic status in children and adolescents (Dariusch Philipp)	Were the two groups similar and recruited from the same population?	Yes	No	Unclear	Not applicable
	Were the exposures measured similarly to assign people to both the exposed and unexposed groups?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the groups/participants free of the outcome at the start of the study (or at the moment of the exposure)?	Yes	No	Unclear	Not applicable
	Were the out comes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was the follow-up time reported and sufficient to be long enough for outcomes to occur?	Yes	No	Unclear	Not applicable
	Was follow-up complete and, if not, were the reasons to loss to follow-up described and explored?	Yes	No	Unclear	Not applicable
	Were strategies to address incomplete follow-up utilized?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
52 Risk Factors for Incident Myopia in Australian Schoolchildren (Amanda N. French, BAppSci (Orth) Hons)	Were the two groups similar and recruited from the same population?	Yes	No	Unclear	Not applicable
	Were the exposures measured similarly to assign people to both the exposed and unexposed groups?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the groups/participants free of the outcome at the start of the study (or at the moment of the exposure)?	Yes	No	Unclear	Not applicable
	Were the out comes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was the follow-up time reported and sufficient to be long enough for outcomes to occur?	Yes	No	Unclear	Not applicable
	Was follow-up complete and, if not, were the reasons to loss to follow-up described and explored?	Yes	No	Unclear	Not applicable

	Were strategies to address incomplete follow-up utilized?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
63 Low physical activity and higher use of screen devices are associated with myopia at the age of 16-17 years in the CCC2000 Eye Study (Mathias Hvidtfelt Hansen)	Were the two groups similar and recruited from the same population?	Yes	No	Unclear	Not applicable
	Were the exposures measured similarly to assign people to both the exposed and unexposed groups?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the groups/participants free of the outcome at the start of the study (or at the moment of the exposure)?	Yes	No	Unclear	Not applicable
	Were the out comes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was the follow-up time reported and sufficient to be long enough for outcomes to occur?	Yes	No	Unclear	Not applicable
	Was follow-up complete and, if not, were the reasons to loss to follow-up described and explored?	Yes	No	Unclear	Not applicable
	Were strategies to address incomplete follow-up utilized?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
70 The Associations between Near Visual Activity and Incident Myopia in Children (Po-Wen Ku)	Were the two groups similar and recruited from the same population?	Yes	No	Unclear	Not applicable
	Were the exposures measured similarly to assign people to both the exposed and unexposed groups?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the groups/participants free of the outcome at the start of the study (or at the moment of the exposure)?	Yes	No	Unclear	Not applicable
	Were the out comes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was the follow-up time reported and sufficient to be long enough for outcomes to occur?	Yes	No	Unclear	Not applicable

	Was follow-up complete and, if not, were the reasons to loss to follow-up described and explored?	Yes	No	Unclear	Not applicable
	Were strategies to address incomplete follow-up utilized?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
71 Risk Factors for Incident Myopia among Teenaged Students ofthe Experimental Class of the Air Force in China (Lin-song Qi)	Were the two groups similar and recruited from the same population?	Yes	No	Unclear	Not applicable
	Were the exposures measured similarly to assign people to both the exposed and unexposed groups?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the groups/participants free of the outcome at the start of the study (or at the moment of the exposure)?	Yes	No	Unclear	Not applicable
	Were the out comes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was the follow-up time reported and sufficient to be long enough for outcomes to occur?	Yes	No	Unclear	Not applicable
	Was follow-up complete and, if not, were the reasons to loss to follow-up described and explored?	Yes	No	Unclear	Not applicable
	Were strategies to address incomplete follow-up utilized?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
78 Physical activity and myopia in Danish children— The CHAMPS Eye Study (Kristian Lundberg)	Were the two groups similar and recruited from the same population?	Yes	No	Unclear	Not applicable
	Were the exposures measured similarly to assign people to both the exposed and unexposed groups?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the groups/participants free of the outcome at the start of the study (or at the moment of the exposure)?	Yes	No	Unclear	Not applicable
	Were the out comes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable

	Was the follow-up time reported and sufficient to be long enough for outcomes to occur?	Yes	No	Unclear	Not applicable
	Was follow-up complete and, if not, were the reasons to loss to follow-up described and explored?	Yes	No	Unclear	Not applicable
	Were strategies to address incomplete follow-up utilized?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Cross-sectional study

Using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for analytical cross-sectional study (last amended in 2023)

Study title (authors)	Major Components	Response options			
1 ASSESSMENT OF THE EXTENT OF MYOPIA AND ITS ASSOCIATED FACTORS IN CHILDREN OF AGE BETWEEN 6 TO 12 YEARS(Dr Mudassar Ali)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
3 Prevalence and risk factors of myopia among children and adolescents in Hangzhou(Lijuan Zhao)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
4 Association between	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable

weekend catch-up outdoor duration and prevalence of myopia: evidence from a cross-sectional, multi-center study in China(Lu Ye)	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
6 Association between weekend catch-up outdoor duration and prevalence of myopia: evidence from a cross-sectional, multi-center study in China(Hanisah Jami Wardati)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
7 Investigation and analysis of the status of myopia and related behavior in children and adolescents in Northeast Sichuan(Wang Ying)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable

	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
8 Regional disparities in the prevalence and correlated factors of myopia in children and adolescents in Gansu, China (Jinyu Wang)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
9 Assessment of potential myopia risk factors, including chronotype, in Estonian adolescents: a cross-sectional study (Teele Palumaa)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
10 The Myopia	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable

Prevalence and Association With Physical Activity Among Primary School Students Aged 6–12 Years: A Cross-Sectional Study in Tianjin, China(Fei Ma)	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
11 Lifestyle and risk of developing myopia in school children in Chongqing, China(Ruili Li)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
12 Nutritional intake, environmental factors, and their impact on myopia prevalence in Korean children aged	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable

5-12 years(Jeong-Mee Kim)	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
14 Prevalence and associated factors of myopia among adolescents aged 12–15 in Shandong Province, China: a cross-sectional study(Zhihao Huang)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
15 Association between myopia and sleep duration among primary school students in minority regions of Southwest China: a school-based cross-sectional research (Ying Huang)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options

17Gender Difference of the Association Between Sleep Duration and Myopia Among Children and Adolescents (Zhaorong Gao)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
18Refractive errors in Tianjin youth aged 6 – 18 years: exploring urban – rural variations and contributing factors (Xin-He Fang)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
20Prevalence and associated factors of myopia among school students in Shenyang, China: a cross-sectional	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable

study（Dan Zhang）	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
26Association of educational environment with the prevalence of myopia: a cross-sectional study in central China （Wei Peng)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
29The association between sleep duration and risk of myopia in Chinese school- aged children: a cross-sectional study （Shujuan Lin)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options

32Prevalence and related factors of children myopia in Pudong New Area, Shanghai: a cross-sectional study (Peng Cheng)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
33Prevalence and related factors of children myopia in Pudong New Area, Shanghai: a cross-sectional study (Peng Cheng)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
34Prevalence and related factors of children myopia in Pudong New Area, Shanghai: a cross-	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable

sectional study (Peng Cheng)	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
36Population-based Evidence of Prevalence and Risk Factors from a Serial Survey in Taiwan (Chiao-Yu Wang)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
40Prevalence of refractive errors and risk factors for myopia among schoolchildren of Almaty, Kazakhstan: A cross-sectional study (Ainagul Mukazhanova)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
45Association between sleep duration and myopia among Chinese children during the COVID-19 pandemic: A cross-sectional study (Luoming Huang)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
46Prevalence and Factors Associated with Myopia Among High School Students in Hawassa City, 2019(South Ethiopia)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
49Reading, writing, working on a computer or watching television, and	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable

myopia(Damian Czepita)	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
55Prevalence and associated factors of myopia among school children in Bahir Dar city, Northwest Ethiopia, 2019 (Abel Sinshaw Assem)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
57Prevalence of myopia and associated risk factors among primary students in Chongqing: multilevel modeling (Zhihao Xie)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
61Factors associated with myopia in Korean children: Korea National Health and nutrition examination survey 2016–2017 (KNHANES VII) (Hyuna Kim)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
62The Prevalence of Myopia and Factors Associated with It Among Secondary School Children in Rural Vietnam(Ho Duc Hung)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
65Ethnic disparity in prevalence and associated	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable

risk factors of myopia in adolescents (Shang-Yi Chiang)	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
68The prevalence and predictors of refractive error among school children in Riyadh, Saudi Arabia (Raed Alomair)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
72Electronic devices and myopic refraction among children aged 6-14 years in urban areas of Tianjin, China (Shengxin Liu)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable

	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
76PREVALENCE OF REFRACTIVE ERRORS AND THE FACTORS RELATED TO IT AMONG MADRASSA STUDENTS IN DISTRICT SIALKOT (Nadia Ashiq)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
77Myopia prevalence and risk factors in children (Christos Theophanous)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
86Prevalence of Myopia and Its Risk Factors in	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable

Urban School Children in Delhi: The North India Myopia Study (NIM Study) (Rohit Saxena)	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
87Risk Factors for Childhood Myopia: Findings From the NICER Study （Lisa O'Donoghue)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
88Prevalence of and factors associated with myopia in primary school students in the Chaoyang District of Beijing, China （Yanyu Lyu)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable

	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable

Study title (authors)	Major Components	Response options			
95Prevalence of refractive errors in teenage high school students in Singapore （Timothy P. L. Quek)	Were the criteria for inclusion in the sample clearly defined?	Yes	No	Unclear	Not applicable
	Were the study subjects and the setting described in detail?	Yes	No	Unclear	Not applicable
	Was the exposure measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Were objective, standard criteria used for measurement of the condition?	Yes	No	Unclear	Not applicable
	Were confounding factors identified?	Yes	No	Unclear	Not applicable
	Were strategies to deal with confounding factors stated?	Yes	No	Unclear	Not applicable
	Were the outcomes measured in a valid and reliable way?	Yes	No	Unclear	Not applicable
	Was appropriate statistical analysis used?	Yes	No	Unclear	Not applicable